EXPLORATORY DATA ANALYSIS

Retail Superstore Sales Dataset

Data Science Internship Program — Task 01

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Internship Domain	Data Science
Task Number	Task 01 — Exploratory Data Analysis
Technologies	Python, Pandas, NumPy, Matplotlib, Seaborn, Google Colab
GitHub Repository	https://github.com/Cedra355

Important Dataset note

The dataset used in this project is "superstore dataset.csv", obtained from Kaggle. The dataset contains retail sales information, including order details, customer information, product categories, sales, profit, discounts, and geographical data. It is used to perform exploratory data analysis (EDA) and identify meaningful business insights.

1. Executive Summary

This project applies exploratory data analysis (EDA) to a retail Superstore dataset to understand sales, profit, product categories, customer segments, regions, discounts and monthly trends. The notebook reports 9,994 order-line records across 21 fields, representing 5,009 unique orders. Total sales are reported as $2,297,200.86 and total profit as $286,397.02. The analysis also identifies low-profit or loss-making areas that merit closer business review.

The main practical message is that sales volume alone is not enough to judge performance: Furniture produces substantial sales but a comparatively low margin, while large discounts and certain sub-categories are associated with losses.

2. Objective

To inspect and clean the dataset, summarize numerical and categorical features, analyze correlations, create at least 12 meaningful visualizations, derive evidence-based business insights, and propose practical recommendations.

3. Dataset Description

Item	Description
Dataset file	Superstore Sale.csv
Records / fields	9,994 rows × 21 columns
Unique orders	5,009 (reported in notebook narrative)
Geographic coverage	United States retail records with region, state and city fields
Main fields	Order and ship dates, ship mode, customer/segment, geography, product category/sub-category, sales, quantity, discount, profit
Dataset source	"Kaggle – Superstore Sales Dataset" (https://www.kaggle.com/datasets/ishanshrivastava28/superstore-sales)

4. Technologies Used

- Python — analysis language.
- Pandas and NumPy — data loading, transformation and statistical summaries.
- Matplotlib and Seaborn — charts and visual exploration.
- Google Colab / Jupyter Notebook — execution environment.

5. Data Import and Understanding

The notebook imports Pandas, NumPy, Matplotlib and Seaborn, uploads the CSV file, reads it using Latin-1 encoding, and examines the first and last rows, shape, column names, data types and descriptive statistics. The data contains 21 columns: 3 floating-point measures (Sales, Discount, Profit), integer identifiers/measures (including Row ID, Postal Code and Quantity), and categorical/text fields.

The notebook prints the number of columns as 21. Its dataset shape is 9,994 rows and 21 columns.

6. Data Cleaning Process

Missing values: The notebook's missing-value check reports 0 null values in every column.

Duplicate records: The notebook reports 0 fully duplicated rows.

Date types: Order Date and Ship Date are converted from text to datetime using pandas.to_datetime(errors='coerce').

Data types: Sales, Quantity, Discount and Profit are numeric in the loaded data; identifiers and descriptive fields are kept as text/integer values as appropriate.

Validation limitation: The notebook checks types and nulls, but it does not show a complete validation of invalid values or a post-conversion count of dates coerced to missing values. These checks should be added before final submission.

7. Feature and Statistical Analysis

The notebook summarizes numerical features with descriptive statistics and adds mean, median, variance and skewness. It also calculates minimum, maximum, mean and median, summarizes categorical values by counts and percentages, plots numerical distributions, and uses box plots plus the interquartile-range (IQR) rule to flag potential outliers.

The main business measures are Sales (transaction-line revenue), Profit (transaction-line profitability), Quantity (units), and Discount (discount rate). Category, Sub-Category, Segment, Region and State help explain where sales and profit originate. Outliers should be investigated rather than automatically removed, because unusually large orders may be legitimate.

A notebook cell contains an additional category-summary snippet using column names such as Product line, Branch, Customer type, Payment and Total; those columns are not present in this Superstore dataset. That snippet does not run its intended analysis for this dataset and should be removed or rewritten to use the actual column names.

8. Correlation Analysis

The notebook calculates correlations for Sales, Quantity, Discount and Profit and visualizes them in a heatmap. Its written findings report the following relationships:

Feature pair	Correlation	Interpretation recorded in notebook
Sales – Profit	0.48	Moderate positive: higher sales tend to accompany higher profit.
Discount – Profit	-0.22	Weak negative: higher discounts are slightly associated with lower profit.
Sales – Quantity	0.20	Weak positive: higher quantities may accompany slightly higher sales.
Sales – Discount	-0.03	Very weak / near-zero linear relationship.
Quantity – Discount	0.01	Very weak / near-zero linear relationship.

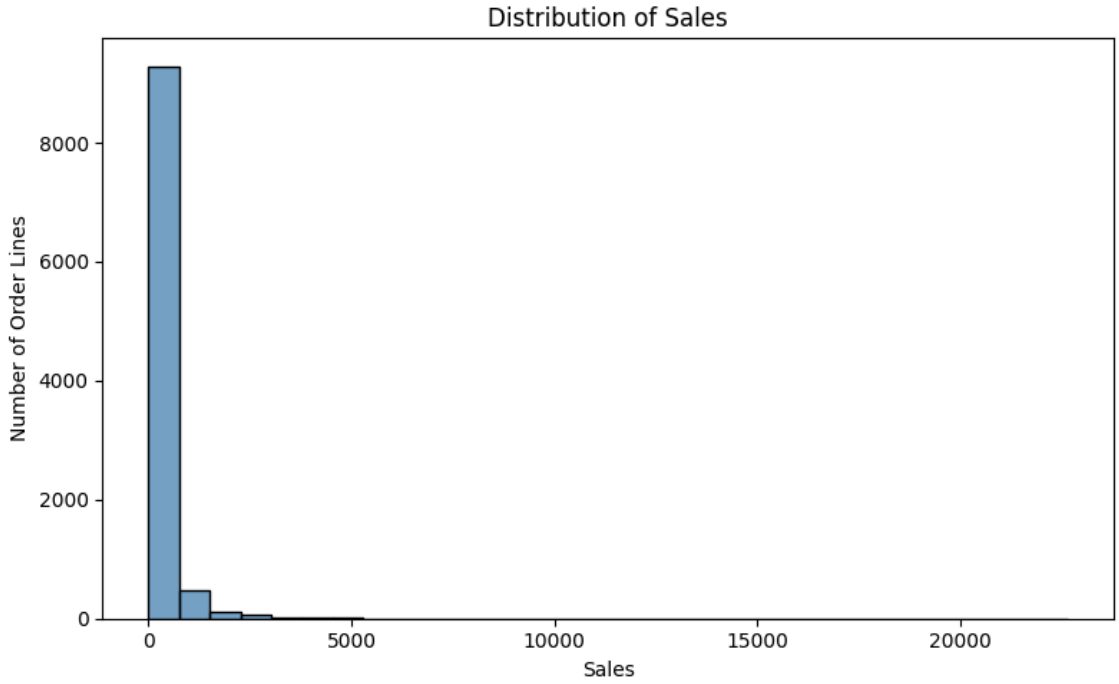
Correlation describes association, not causation. For example, the negative Discount–Profit relationship does not alone prove that discounts cause losses; product mix, costs, and other factors may also matter.

9. Data Visualizations

The following 12 figures are embedded from the submitted notebook outputs, not recreated from an assumed dataset. Each figure is paired with its intended analytical purpose.

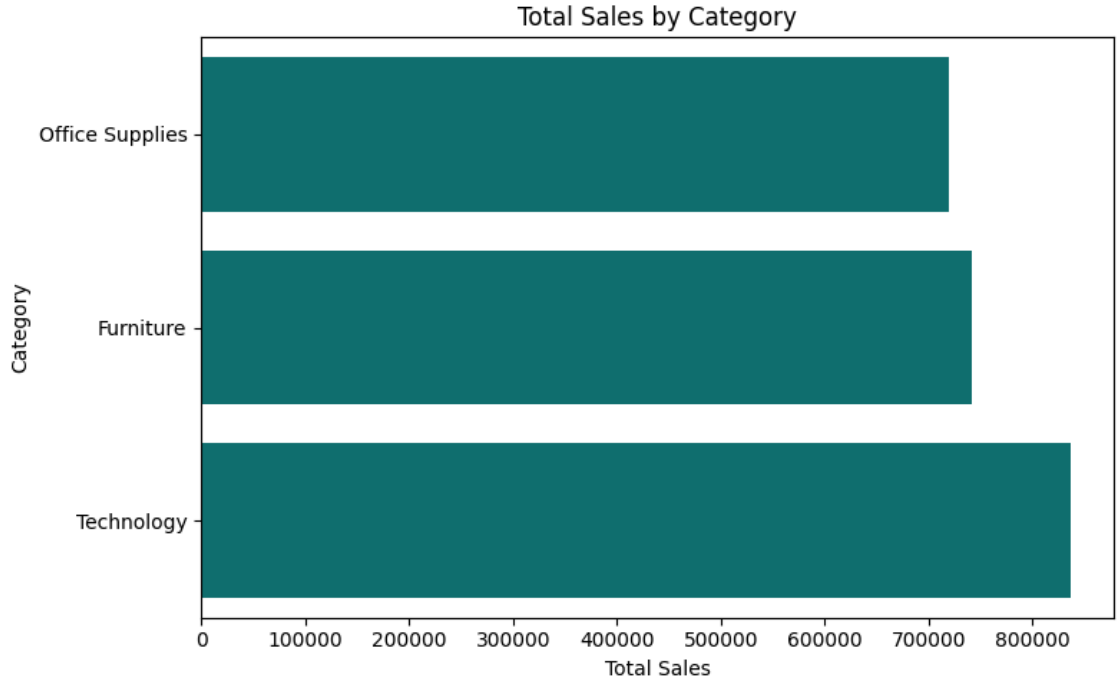
Visualization 1 — Sales Distribution

Sales are concentrated at lower transaction values, with a long right tail.



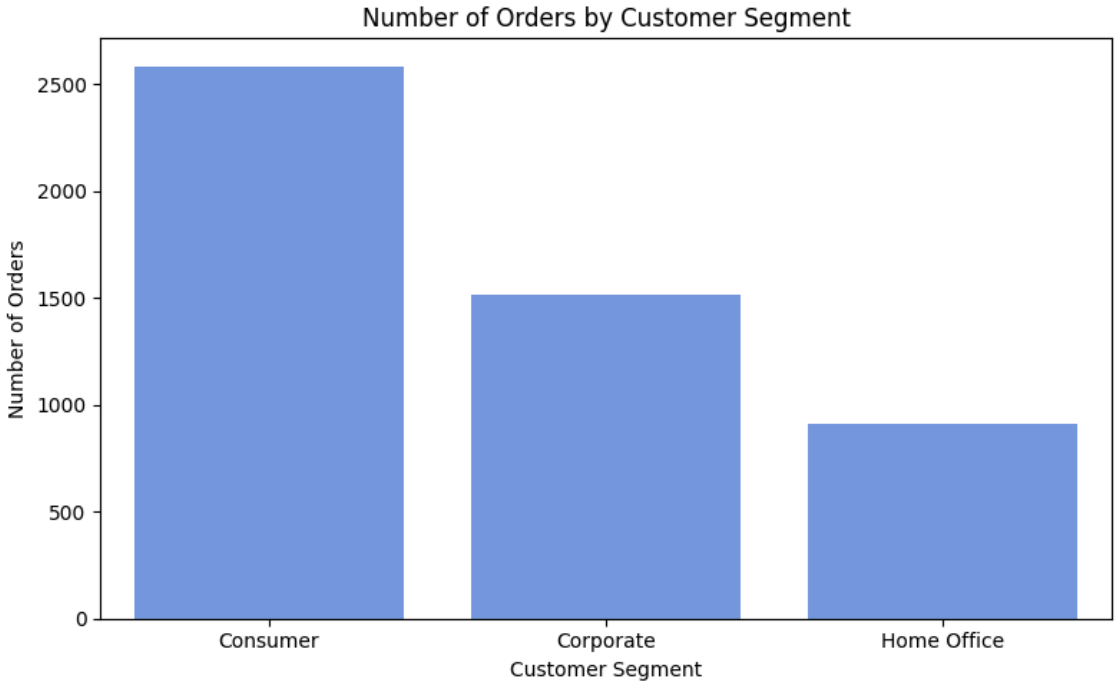
Visualization 2 — Total Sales by Category

Compares total sales across the three product categories.



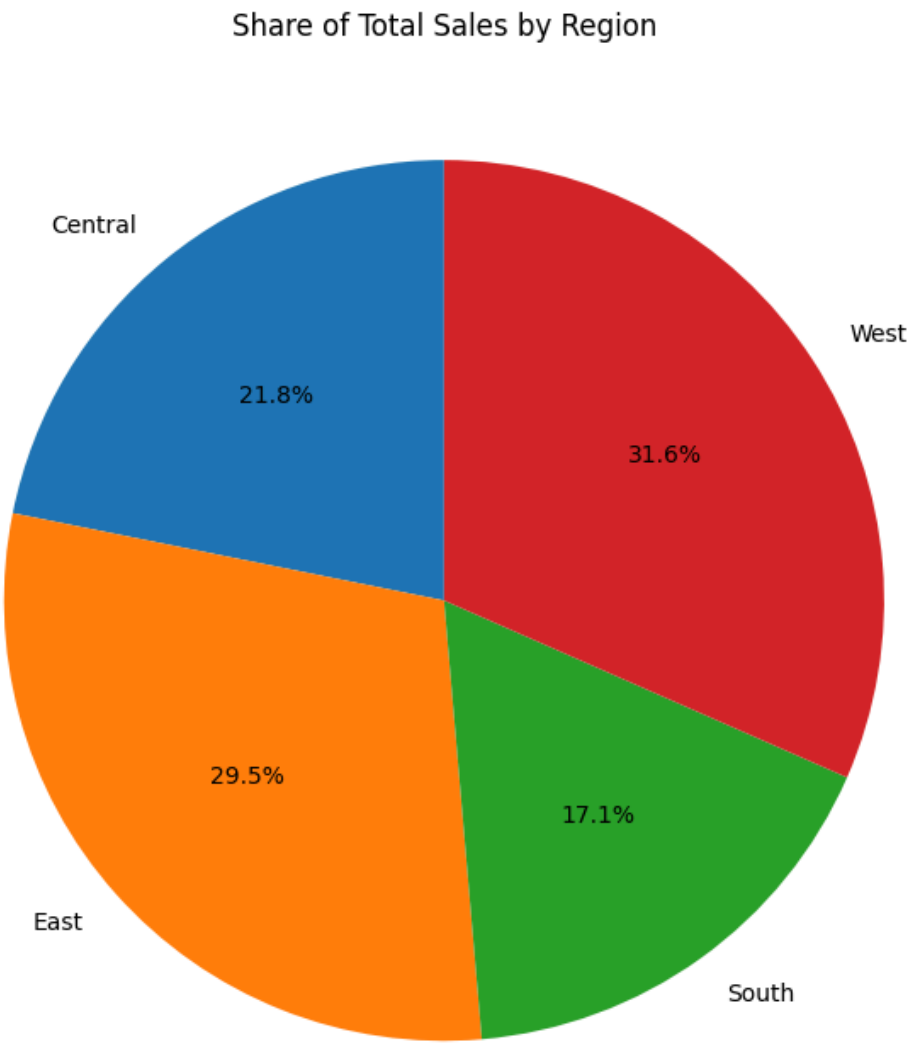
Visualization 3 — Orders by Customer Segment

Counts unique orders by customer segment.



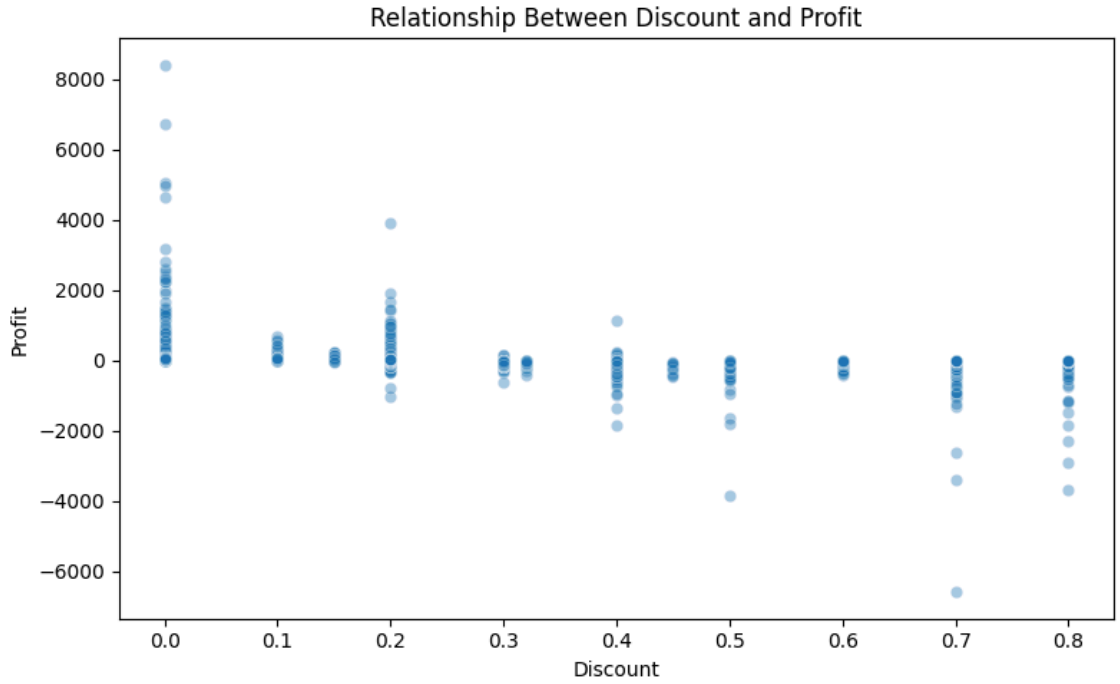
Visualization 4 — Sales Share by Region

Shows each region's contribution to total sales.



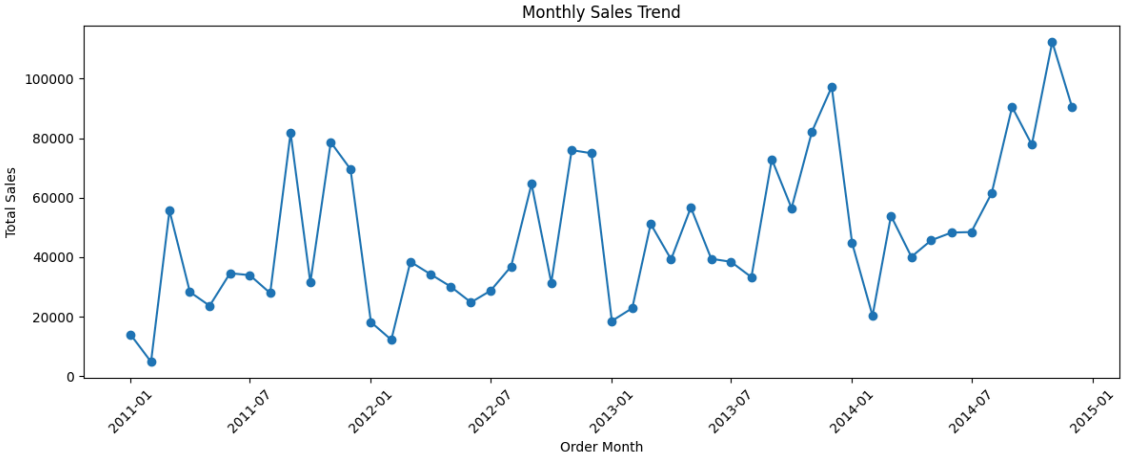
Visualization 5 — Discount vs Profit

Shows the spread of profit outcomes at different discount levels.



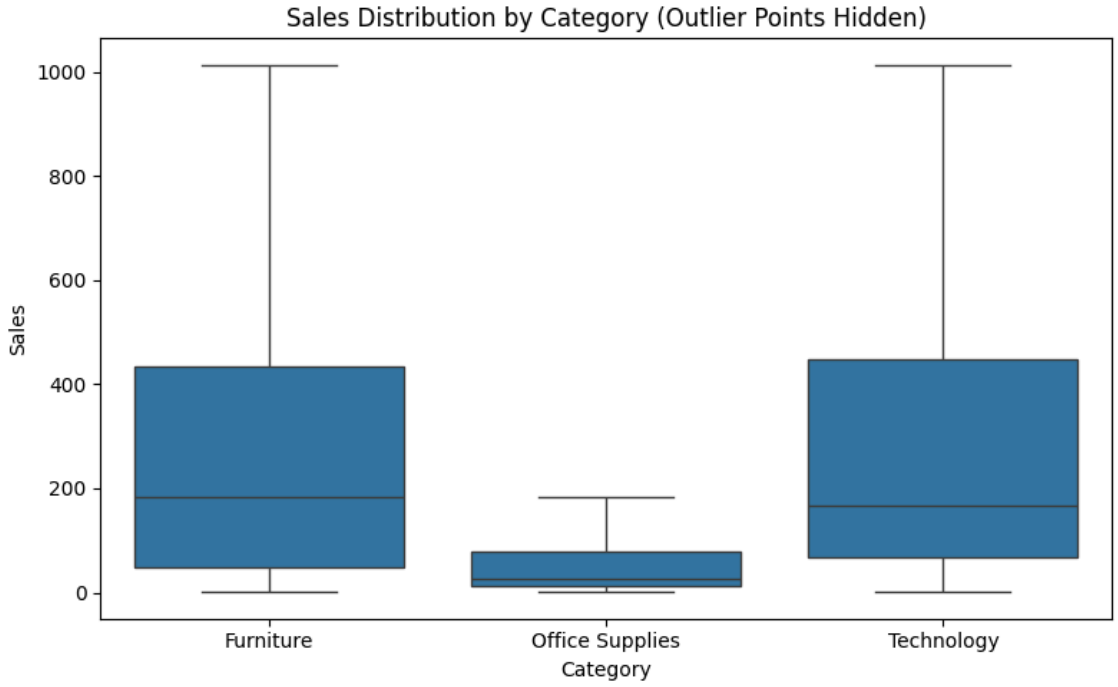
Visualization 6 — Monthly Sales Trend

Shows monthly sales over the date range available in the dataset.



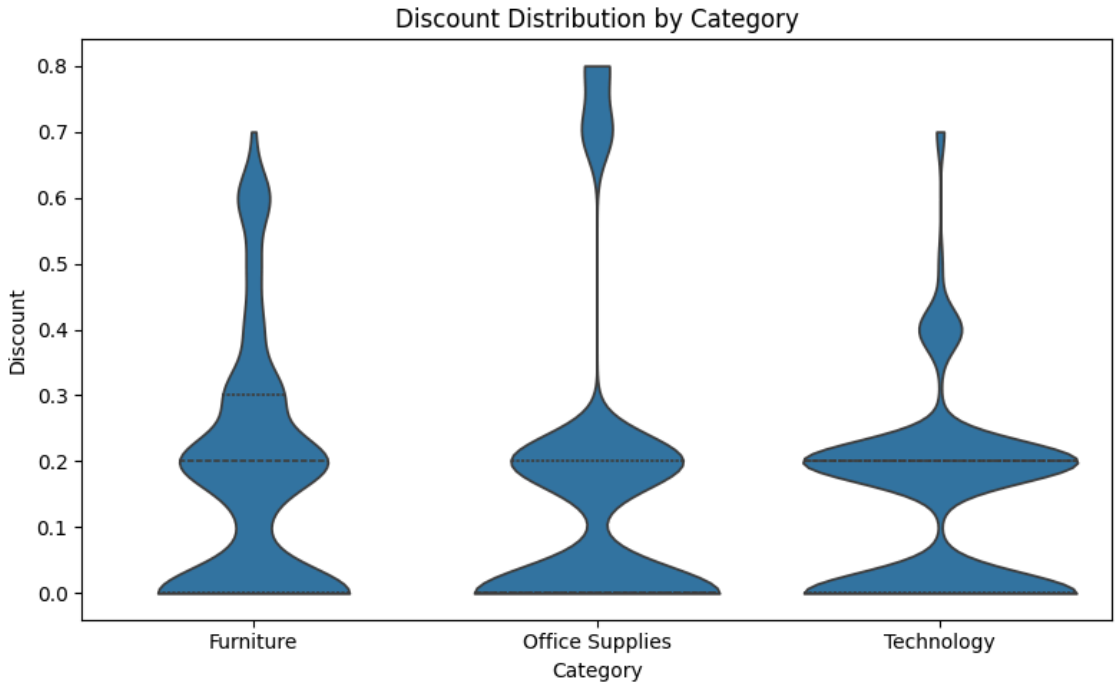
Visualization 7 — Sales Distribution by Category

Compares category-level sales distributions; outlier points are hidden for readability.



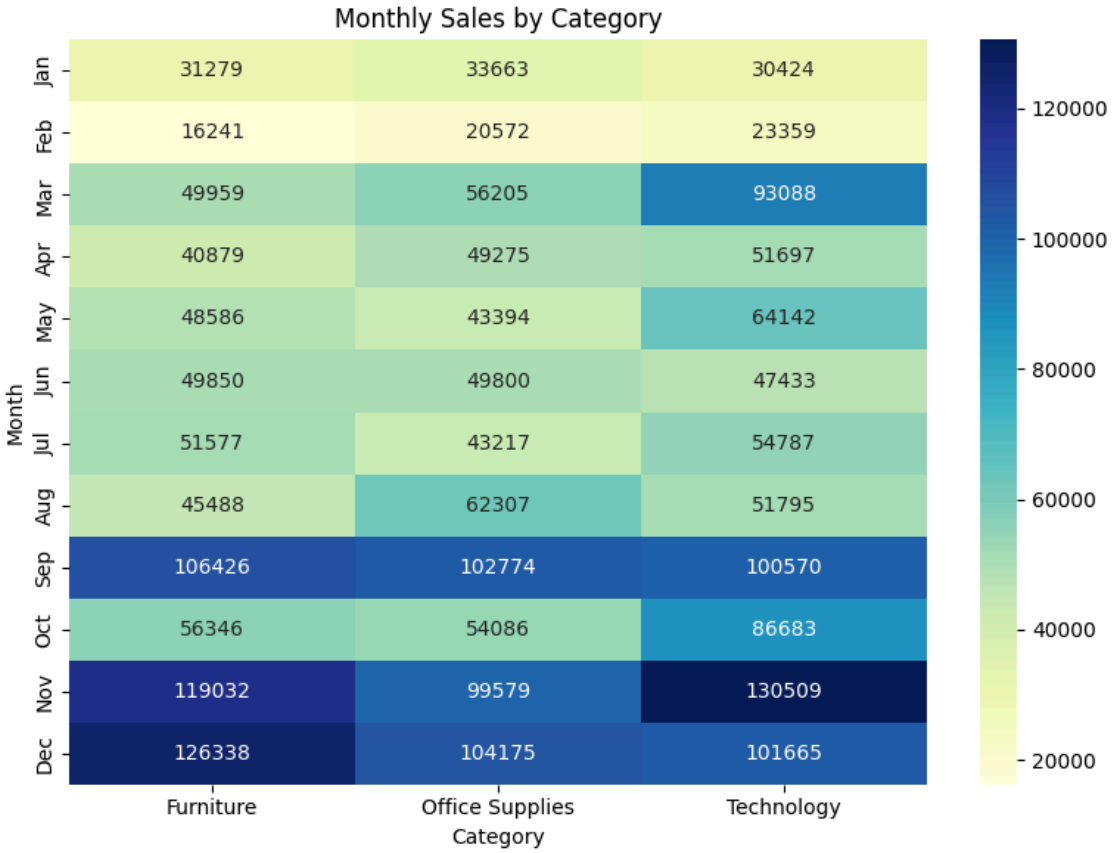
Visualization 8 — Discount Distribution by Category

Compares discount distributions across product categories.



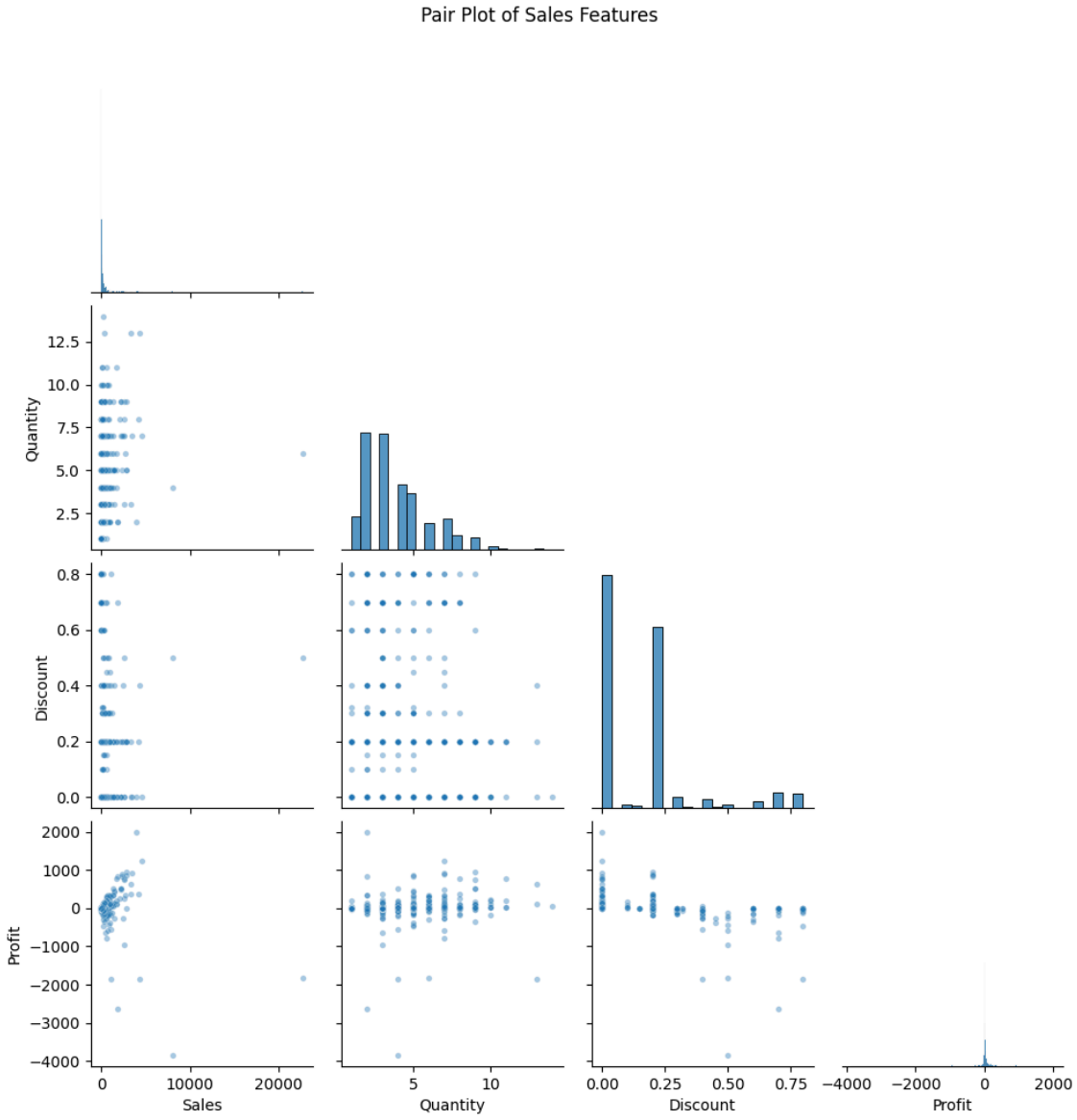
Visualization 9 — Monthly Sales by Category Heatmap

Highlights monthly sales variation by category.



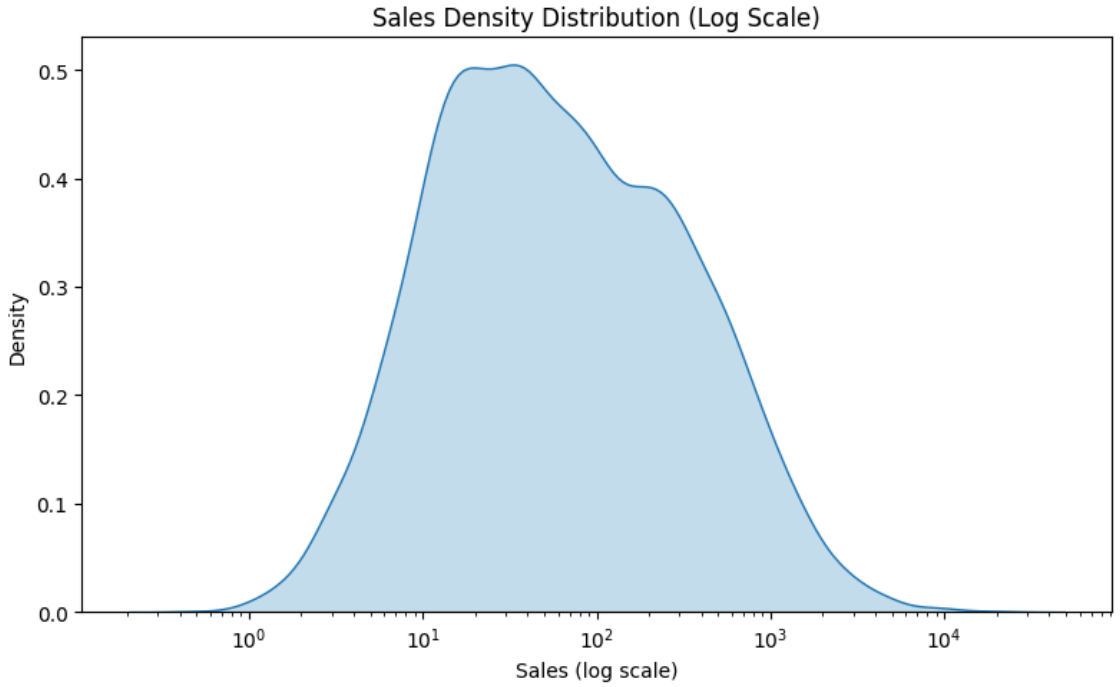
Visualization 10 — Pair Plot of Numeric Features

Explores pairwise patterns among sales, quantity, discount and profit.



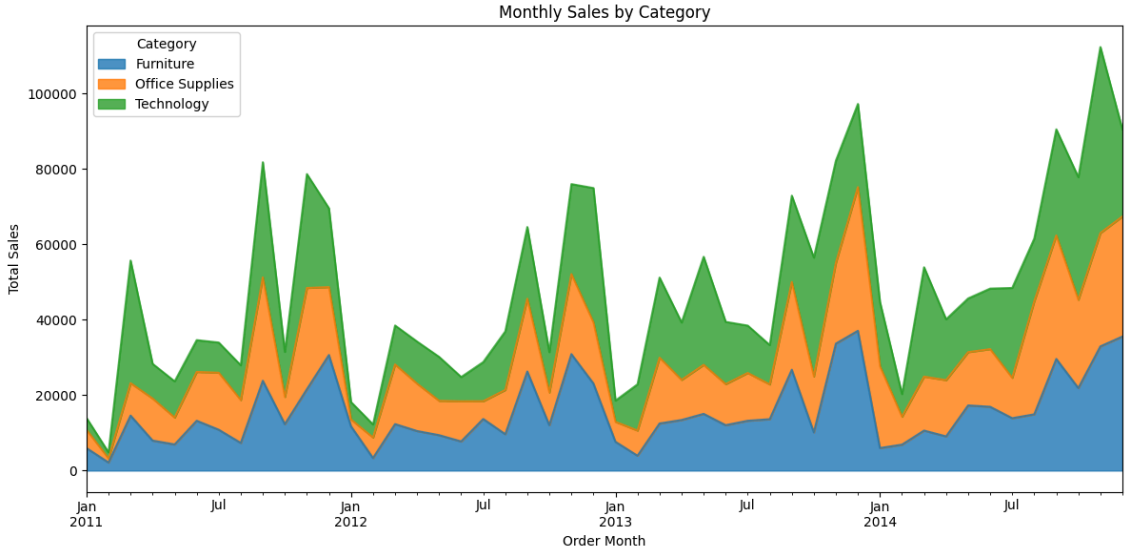
Visualization 11 — Sales Density (Log Scale)

Shows the distribution of sales on a logarithmic scale.



Visualization 12 — Monthly Sales by Category Area Chart

Shows how category sales contribute to the monthly total.



10. Business Insights

The following insights preserve the findings written in the notebook. Currency values and interpretations below are reported by the submitted analysis and should be validated against the original CSV before external publication.

1. Overall performance: 9,994 order-line records and 5,009 unique orders generated reported sales of $2,297,200.86 and profit of $286,397.02.
2. Technology leads sales and profit: it generated $836,154.03 in sales and $145,454.95 in profit, the highest among the three categories.
3. Office Supplies contributes substantial profit: reported sales were $719,047.03 and profit was $122,490.80.
4. Furniture has a comparatively low profit margin: it generated $741,999.80 sales but only $18,451.27 profit, approximately a 2.5% margin.
5. Consumers account for the largest sales share: the Consumer segment generated $1,161,401.34, about 50.6% of total sales.
6. Home Office has the highest segment profit margin: reported profit was $60,298.68 on sales of $429,653.15, around 14% margin.
7. The West is the top region by sales and profit: reported sales were $725,457.82 and profit was $108,418.45.
8. Central has the lowest regional profit margin: sales were $501,239.89, profit was $39,706.36, margin about 7.9%; the notebook also reports an average discount of 24%.
9. Texas combines high sales with the largest state-level loss: reported sales were $170,188.05 and loss was $25,729.36.
10. Tables is the largest loss-making sub-category: sales were $206,965.53 and loss was $17,725.48.
11. Bookcases and Supplies also recorded losses: reported losses were $3,472.56 and $1,189.10, respectively.
12. Copiers had the highest sub-category profit: $55,617.82 profit from $149,528.03 sales.
13. Phones and Chairs lead sub-category sales: Phones generated $330,007.05 sales and Chairs generated $328,449.10.
14. Loss-making order lines are a significant minority: 1,871 lines (reported as 18.7% of records) were loss-making, with total losses of $156,131.29.
15. No-discount lines had higher average profit than 20%-discount lines: average profit was reported as $66.90 with no discount versus $24.70 at a 20% discount.
16. The 30% discount group frequently recorded losses: among 227 order lines with a 30% discount, 91.6% were loss-making; average profit was −$45.68.
17. Fourth-quarter sales were highest: the notebook reports Q4 sales of $368,948.62 compared with Q1 sales of $116,661.29. The quarterly aggregation method should be checked before using these values.
18. Sales rose while profit fell from 2013 to 2014: sales increased from $261,514.21 to $280,923.61, while profit decreased from $43,667.09 to $27,952.92.

11. Business Improvement Recommendations

Review the discount policy. The notebook reports that 91.6% of order lines with a 30% discount made a loss. Review high discounts and require margin checks or approval for steep discounts rather than applying them broadly.

Improve Furniture profitability. Furniture has high sales but a reported profit margin of only about 2.5%. Examine product costs, shipping costs and discount levels to identify the drivers of low margin.

Focus on loss-making sub-categories. Tables, Bookcases and Supplies were reported as loss-making. Review pricing, cost structure, promotions and product mix before continuing the same sales strategy.

Investigate Texas and Central region performance. Texas had the largest state-level loss in the notebook, while Central had the lowest regional profit margin. Break down discounting, category mix and fulfilment costs by state and product.

Promote profitable products. Copiers, Phones, Accessories and Paper were identified as strong profit contributors. Test targeted campaigns, bundles and cross-selling while monitoring incremental profit, not sales alone.

Use segment-specific offers. Consumer customers contributed the largest sales total, while Home Office had the highest reported profit margin. Tailor offers to each segment and measure profit per order.

Plan for fourth-quarter demand. The notebook identifies Q4 as the strongest sales quarter. Prepare inventory, staffing and promotions ahead of peak demand, after validating the quarter-level aggregation.

12. Conclusion

The EDA workflow covers dataset loading, basic inspection, null and duplicate checks, date conversion, descriptive statistics, outlier exploration, correlation analysis and 12 visualizations. The written findings point to strong Technology and West-region performance, weaker Furniture profitability, loss-making sub-categories, and a need to review steep discounts. These findings support practical next steps around margin management, regional diagnosis, product-level decisions and seasonal planning.

Before submission, the notebook should be checked for a few quality issues: (1) its filename and schema indicate Superstore rather than Supermarket Sales; (2) one feature-analysis cell references columns not found in this dataset and includes a typo in an aggregation function; (3) verify the date parsing and count any dates coerced to missing; (4) confirm quarterly sales calculations; and (5) add a real GitHub repository URL and the student name on the cover.